

A SOURCE-ANCHORED ACKERMANN COMPARISON FOR THE SEIDEL–SHARIR TOP-DOWN PATH-COMPRESSION HIERARCHY

PATH-COMPRESSSION PROOF-DIGESTION PACKET

ABSTRACT. This exposition packages the proof that the Seidel–Sharir top-down J hierarchy implies an inverse-Ackermann bound under the packet normalization. The main comparison is

$$R_{z+1}(Q) \geq A(z, 4Q) \quad (z \geq 1, Q \geq 1),$$

with constants $c = 1$, $C = 1$, and $D = 4$ in the comparison template $R_{cz+C}(Q) \geq A(z, DQ)$. The note is a source-fidelity and threshold-inversion exposition. It is not a new union-find bound, and it is not a replacement for the full Seidel–Sharir recurrence proof.

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1. INTRODUCTION

Tarjan’s Dagstuhl prompt asks for a simple direct proof that the Seidel–Sharir top-down J hierarchy for path compression implies an inverse-Ackermann bound using a classical Ackermann definition; see the problem anchor in the Dagstuhl report and the original Seidel–Sharir paper [2, 1].

This note proves that comparison under the packet normalization. It uses the source J hierarchy and the packet Ackermann function to prove

$$R_{z+1}(Q) \geq A(z, 4Q) \quad (z \geq 1, Q \geq 1).$$

Thus the packet comparison constants are explicit:

$$c = 1, \quad C = 1, \quad D = 4.$$

The role of this document is deliberately narrow. It does not claim a new union-find bound, does not reprove the full Seidel–Sharir top-down recurrence, and does not replace that recurrence with a bottom-up Tarjan-potential argument. The only recurrence used here is the source-anchored consequence, where $f(m, n, r)$ denotes the path-compression cost function in the Seidel–Sharir recurrence [1, p. 9],

$$f(m, n, r) \leq km + 2nJ_k(r).$$

2. SOURCE RECURRENCE AND CONVENTIONS

All logarithms are base 2. The source definitions and recurrence quoted in this section are from Seidel–Sharir [1, pp. 7–9]. For an integer-valued function g with $g(r) < r$ for positive r , the source defines

$$g^*(r) = \begin{cases} 0, & r \leq 1, \\ 1 + g^*(g(r)), & r > 1, \end{cases}$$

and

$$g^\diamond(r) = \begin{cases} g(r), & g(r) \leq 1, \\ 1 + g^\diamond(\lceil \log_2 g(r) \rceil), & g(r) > 1. \end{cases}$$

The source shifting lemma, with the $2n$ coefficient used below, is Lemma 5 of Seidel–Sharir [1, pp. 7–8].

Lemma 2.1 (Source-correct shifting lemma). *If*

$$f(m, n, r) \leq km + 2ng(r),$$

then

$$f(m, n, r) \leq (k + 1)m + 2ng^\diamond(r).$$

The source hierarchy is Corollary 6 of Seidel–Sharir [1, p. 9]:

$$J_0(r) = \left\lceil \frac{r-1}{2} \right\rceil, \quad J_{k+1} = J_k^\diamond.$$

Equivalently,

$$J_{k+1}(r) = \begin{cases} J_k(r), & J_k(r) \leq 1, \\ 1 + J_{k+1}(\lceil \log_2 J_k(r) \rceil), & J_k(r) > 1. \end{cases}$$

The same corollary gives the source recurrence consequence used in the cost bound:

$$f(m, n, r) \leq km + 2nJ_k(r).$$

3. THRESHOLD INVERSION

For $t \geq 0$, define the threshold inverse

$$R_k(t) = \max\{r \geq 0 : J_k(r) \leq t\}.$$

The following package supplies the finite-threshold and unboundedness facts needed for this definition.

Lemma 3.1 (Basic J package). *For every $k \geq 0$:*

- (1) J_k is integer-valued, nonnegative, nondecreasing, and unbounded on $\mathbb{N}$.
- (2) $J_k(0) = 0$ and $J_k(r) < r$ for every $r > 1$.
- (3) $J_{k+1}(r) \leq J_k(r)$ for every $r \geq 0$.
- (4) $R_k(t)$ is finite for every finite $t \geq 0$.
- (5) $R_k(t)$ is nondecreasing in t , and $R_{k+1}(t) \geq R_k(t)$.

Proof. For $k = 0$, the formula $J_0(r) = \lceil (r-1)/2 \rceil$ gives the asserted properties directly. For the induction step, let $h = g^\diamond$ where g is integer-valued, nonnegative, nondecreasing, unbounded, satisfies $g(0) = 0$, and $g(r) < r$ for $r > 1$. Since

$$\lceil \log_2 x \rceil \leq x - 1 \quad (x \geq 2),$$

the recursive branch of h calls h only on smaller arguments. A strong induction on r gives $h(r) \leq g(r)$ for all r , hence $h(r) < r$ for $r > 1$ and $h(0) = 0$. Monotonicity follows by comparing $g(r) \leq g(s)$ and, in the recursive branch, using the monotonicity of the ceiling logarithm and the strong induction hypothesis. To see unboundedness, if $h(u) \geq M$ and $g(r) > 2^u$, then

$$h(r) = 1 + h(\lceil \log_2 g(r) \rceil) \geq M + 1.$$

Thus $g^\diamond$ preserves the stated properties, and induction gives the claims for every J_k .

Since each J_k is nondecreasing and unbounded, the set $\{r \geq 0 : J_k(r) \leq t\}$ is nonempty and finite for finite $t \geq 0$. This proves finiteness of $R_k(t)$. Monotonicity in t is immediate from inclusion of threshold sets, and $J_{k+1} \leq J_k$ gives $R_{k+1}(t) \geq R_k(t)$. $\square$

Lemma 3.2 (Exact base inverse). *For every $t \geq 0$,*

$$R_0(t) = 2t + 1.$$

Proof. The inequality $J_0(r) \leq t$ is equivalent to $\lceil (r-1)/2 \rceil \leq t$, hence to $r \leq 2t + 1$. $\square$

Lemma 3.3 (Threshold step). *Let B, t be nonnegative integers. If*

$$r \leq R_k(2^B) \quad \text{and} \quad B \leq R_{k+1}(t),$$

then

$$J_{k+1}(r) \leq t + 1.$$

Consequently,

$$R_{k+1}(t + 1) \geq R_k(2^{R_{k+1}(t)}).$$

Proof. The first hypothesis gives $J_k(r) \leq 2^B$. If $J_k(r) \leq 1$, then $J_{k+1}(r) = J_k(r) \leq 1 \leq t + 1$. If $J_k(r) > 1$, then

$$J_{k+1}(r) = 1 + J_{k+1}(\lceil \log_2 J_k(r) \rceil),$$

and $\lceil \log_2 J_k(r) \rceil \leq B$. Since J_{k+1} is nondecreasing and $B \leq R_{k+1}(t)$,

$$J_{k+1}(\lceil \log_2 J_k(r) \rceil) \leq J_{k+1}(B) \leq t.$$

Thus $J_{k+1}(r) \leq t + 1$. Taking $B = R_{k+1}(t)$ gives the displayed threshold recurrence. $\square$

Lemma 3.4 (Threshold jump). *For every $k \geq 0$ and every $Q \geq 2$,*

$$R_{k+1}(Q) \geq R_k(4Q).$$

Proof. Apply Lemma 3.3 with $t = Q - 1$:

$$R_{k+1}(Q) \geq R_k(2^{R_{k+1}(Q-1)}).$$

The basic package and the exact base inverse give $R_{k+1}(Q-1) \geq R_0(Q-1) = 2Q - 1$. For $Q \geq 2$, $2^{2Q-1} \geq 4Q$, and monotonicity of R_k finishes the proof. $\square$

4. ACKERMANN COMPARISON

The packet Ackermann function is defined exactly as follows:

$$A(0, x) = 2x,$$

and for $i \geq 1$,

$$A(i, 0) = 0, \quad A(i, 1) = 2, \quad A(i, x) = A(i - 1, A(i, x - 1)) \quad (x \geq 2).$$

In particular, $A(1, x) = 2^x$ for $x \geq 1$.

Lemma 4.1 (Ackermann monotonicity and row domination). *For the packet Ackermann function:*

- (1) *for every $i \geq 0$, $A(i, x)$ is nondecreasing in x ;*
- (2) *for every $i \geq 1$ and $x \geq 1$, $A(i, x) \geq 2x$;*
- (3) *for every $i \geq 0$ and $x \geq 1$, $A(i + 1, x) \geq A(i, x)$.*

Proof. For $i = 0$, monotonicity follows from $A(0, x) = 2x$. For $i \geq 1$, monotonicity follows inductively from $A(i, x + 1) = A(i - 1, A(i, x))$ and the lower-row monotonicity.

The growth bound is by induction on x . At $x = 1$ it says $2 = 2x$. For $x \geq 2$,

$$A(i, x) = A(i - 1, A(i, x - 1)).$$

If $i = 1$, this equals $2A(1, x - 1)$; if $i \geq 2$, the lower-row growth bound gives at least $2A(i, x - 1)$. Hence $A(i, x) \geq 2x$.

For row domination, equality holds at $x = 1$. For $x \geq 2$,

$$A(i + 1, x) = A(i, A(i + 1, x - 1)).$$

The growth bound gives $A(i + 1, x - 1) \geq 2(x - 1) \geq x$, and monotonicity in row i gives $A(i + 1, x) \geq A(i, x)$. $\square$

Lemma 4.2 (Row-domination invariant). *For every $j \geq 1$ and $x \geq 1$,*

$$R_j(x) \geq A(j+1, x).$$

Proof. We prove this by induction on j , with an inner induction on x .

For $j = 1$ and $x = 1$, Lemma 3.3 with $k = 0, t = 0$ gives

$$R_1(1) \geq R_0(2^{R_1(0)}).$$

Since $R_1(0) \geq R_0(0) = 1$, this gives $R_1(1) \geq R_0(2) = 5 \geq 2 = A(2, 1)$. If $R_1(x) \geq A(2, x)$, then

$$R_1(x+1) \geq R_0(2^{R_1(x)}) = 2 \cdot 2^{R_1(x)} + 1 \geq 2^{A(2, x)} = A(2, x+1).$$

Now assume $R_j(y) \geq A(j+1, y)$ for all $y \geq 1$. We prove the claim for $j+1$. At $x = 1$, $R_{j+1}(1) \geq R_0(1) = 3 \geq 2 = A(j+2, 1)$. If $R_{j+1}(x) \geq A(j+2, x)$, then Lemma 3.3 gives

$$R_{j+1}(x+1) \geq R_j(2^{R_{j+1}(x)}).$$

By the outer induction hypothesis and Ackermann monotonicity,

$$R_j(2^{R_{j+1}(x)}) \geq A(j+1, 2^{R_{j+1}(x)}) \geq A(j+1, A(j+2, x)).$$

The last expression is $A(j+2, x+1)$ by the Ackermann recursion. $\square$

Theorem 4.3 (Main comparison). *For all integers $z \geq 1$ and $Q \geq 1$,*

$$R_{z+1}(Q) \geq A(z, 4Q).$$

Consequently,

$$A(z, 4Q) > r \implies J_{z+1}(r) \leq Q.$$

Proof. First suppose $Q \geq 2$. By Lemma 3.4, $R_{z+1}(Q) \geq R_z(4Q)$. Since $z \geq 1$, Lemma 4.2 gives $R_z(4Q) \geq A(z+1, 4Q)$. Row domination from Lemma 4.1 gives $A(z+1, 4Q) \geq A(z, 4Q)$.

For $Q = 1, z = 1$, Lemma 3.3 with $k = 1, t = 0$ gives $R_2(1) \geq R_1(2^{R_2(0)})$. Since $R_2(0) \geq R_0(0) = 1$, monotonicity gives $R_2(1) \geq R_1(2)$. The same lemma with $k = 0, t = 0$ gives $R_1(1) \geq R_0(2^{R_1(0)}) \geq R_0(2) = 5$. Applying it again with $k = 0, t = 1$ gives $R_1(2) \geq R_0(2^{R_1(1)}) \geq R_0(2^5) = 65 \geq 16 = A(1, 4)$.

For $Q = 1, z \geq 2$, Lemma 3.3 with $k = z, t = 0$ gives $R_{z+1}(1) \geq R_z(2^{R_{z+1}(0)})$. Since $R_{z+1}(0) \geq R_0(0) = 1$, monotonicity gives $R_{z+1}(1) \geq R_z(2)$. Applying Lemma 3.4 with level $z-1$ and $Q = 2$ gives $R_z(2) \geq R_{z-1}(8)$. Since $z-1 \geq 1$, Lemma 4.2 gives $R_{z-1}(8) \geq A(z, 8) \geq A(z, 4)$.

Thus $R_{z+1}(Q) \geq A(z, 4Q)$ in all cases. If $A(z, 4Q) > r$, then $r \leq A(z, 4Q) - 1 \leq R_{z+1}(Q)$, so $J_{z+1}(r) \leq Q$ by the definition of R_{z+1} . $\square$

5. PATH-COMPRESSION CONSEQUENCE

Define

$$L(n) = \lceil \log_2 \max(n, 2) \rceil, \quad Q(m, n) = \left\lceil 1 + \frac{m}{n} \right\rceil.$$

Define

$$\alpha_Q(m, n) = \min\{z \geq 1 : A(z, 4Q(m, n)) > L(n)\},$$

$$\alpha_J^Q(m, n) = \min\{k \geq 0 : J_k(L(n)) \leq Q(m, n)\},$$

and

$$\alpha_J^S(m, n) = \min\left\{k \geq 0 : J_k(L(n)) \leq 1 + \frac{m}{n}\right\}.$$

Lemma 5.1 (Existence of α_Q). *For positive integers m, n , the minimum defining $\alpha_Q(m, n)$ exists.*

Proof. Since $Q(m, n) \geq 1$, it suffices to prove that $A(z, 4)$ is unbounded in z . Let $a_z = A(z, 4)$. Then $a_1 = 16$, and for $z \geq 1$,

$$A(z+1, 2) = 4, \quad A(z+1, 3) = A(z, 4) = a_z,$$

so $a_{z+1} = A(z+1, 4) = A(z, a_z)$. By Lemma 4.1, $A(z, a_z) \geq 2a_z$. Hence a_z is unbounded, and monotonicity in the second argument gives $A(z, 4Q(m, n)) \geq A(z, 4)$. $\square$

Theorem 5.2 (Packet threshold comparison). *For all positive integers m, n ,*

$$\alpha_J^Q(m, n) \leq \alpha_Q(m, n) + 1.$$

Proof. Let $z = \alpha_Q(m, n)$, $Q = Q(m, n)$, and $L = L(n)$. By definition, $A(z, 4Q) > L$. Theorem 4.3 gives $R_{z+1}(Q) \geq A(z, 4Q) > L$. Since L is an integer, $L \leq R_{z+1}(Q)$, so $J_{z+1}(L) \leq Q$. Therefore $\alpha_J^Q(m, n) \leq z + 1 = \alpha_Q(m, n) + 1$. $\square$

Lemma 5.3 (Ackermann buffer). *For $z \geq 1$ and integer $p \geq 1$,*

$$A(z+1, 4p) \geq A(z, 4p+4).$$

Proof. Since $4p \geq 2$,

$$A(z+1, 4p) = A(z, A(z+1, 4p-1)).$$

If $p = 1$, then $A(z+1, 3) = A(z, 4) \geq 8 = 4p + 4$. If $p \geq 2$, then Lemma 4.1 gives $A(z+1, 4p-1) \geq 2(4p-1) = 8p-2 \geq 4p+4$. Monotonicity of $A(z, -)$ finishes the proof. $\square$

Theorem 5.4 (Source-faithful real threshold). *For all positive integers m, n ,*

$$\alpha_J^S(m, n) \leq \alpha_Q(m, n) + 2.$$

If $1 + m/n$ is integral, then

$$\alpha_J^S(m, n) \leq \alpha_Q(m, n) + 1.$$

Proof. Let $s = 1 + m/n$, $Q = \lceil s \rceil$, $L = L(n)$, and $z = \alpha_Q(m, n)$. Then $A(z, 4Q) > L$.

If s is integral, then $Q = s$. Theorem 4.3 gives $R_{z+1}(s) \geq A(z, 4Q) > L$, hence $J_{z+1}(L) \leq s$. Thus $\alpha_J^S(m, n) \leq z + 1$.

If s is not integral, put $p = \lfloor s \rfloor$. Since $m, n \geq 1$, $p \geq 1$, and $Q = p + 1$. Thus $A(z, 4p + 4) > L$. By Lemma 5.3, $A(z + 1, 4p) > L$. Theorem 4.3 gives $R_{z+2}(p) \geq A(z + 1, 4p) > L$, so $J_{z+2}(L) \leq p < s$. Therefore $\alpha_J^S(m, n) \leq z + 2$. $\square$

Theorem 5.5 (Cost consequence). *Using the source recurrence*

$$f(m, n, r) \leq km + 2nJ_k(r),$$

one obtains

$$f(m, n, L(n)) \leq (\alpha_Q(m, n) + 3)m + 4n.$$

Consequently,

$$f(m, n, L(n)) = O(m\alpha_Q(m, n) + n).$$

Proof. Let $z = \alpha_Q(m, n)$, $Q = Q(m, n)$, and $L = L(n)$. By Theorem 5.2, $J_{z+1}(L) \leq Q$. Apply the source recurrence with $k = z + 1$:

$$f(m, n, L) \leq (z + 1)m + 2nJ_{z+1}(L) \leq (z + 1)m + 2nQ.$$

Since

$$Q = \left\lceil 1 + \frac{m}{n} \right\rceil \leq 2 + \frac{m}{n},$$

we have $2nQ \leq 4n + 2m$. Therefore

$$f(m, n, L(n)) \leq (\alpha_Q(m, n) + 3)m + 4n.$$

The big- O statement follows immediately. $\square$

6. NORMALIZATION AND SOURCE-FIDELITY COMMENTS

The packet integer threshold

$$Q(m, n) = \left\lceil 1 + \frac{m}{n} \right\rceil$$

is not an original source threshold. It is a normalization that lets the integer-valued inverse $R_k(t)$ compare directly with the source real threshold $1 + m/n$ [1, p. 9]. The exact real threshold is handled by Theorem 5.4; the nonintegral case costs one extra level.

The source page 10 comparison with Tarjan's α_T [1, p. 10] is context, not a proof dependency here. The proof above uses only the source J hierarchy, the source shifting lemma and recurrence, the packet Ackermann normalization, and the packet thresholds $L(n)$ and $Q(m, n)$.

The main hazards avoided are:

- the wrong shifting-lemma factor: the source-correct premise is $f(m, n, r) \leq km + 2ng(r)$;
- a silent replacement of the source threshold $1 + m/n$ by $2m/n$;
- drift into a bottom-up Tarjan-potential proof;

- fake equivalence wording: Theorem 4.3 gives the displayed implication, not an unwarranted equivalence.

APPENDIX A. DEPENDENCY TABLE AND DAG

Stage	Input	Output
Source definitions	$g^*, g^\diamond, J_0, J_{k+1} = J_k^\diamond$	The J hierarchy
Source recurrence	Shifting lemma with $2ng(r)$ premise	$f(m, n, r) \leq km + 2nJ_k(r)$
Basic J lemmas	Diamond preservation	J_k monotone, unbounded, $J_k(r) < r$, finite R_k
Threshold recurrence	$J_{k+1} = J_k^\diamond$	$R_{k+1}(t+1) \geq R_k(2^{R_{k+1}(t)})$
Ackermann domination	Threshold recurrence plus $R_0(t) = 2t + 1$	$R_j(x) \geq A(j+1, x)$
Main comparison	Threshold jump and row domination	$R_{z+1}(Q) \geq A(z, 4Q)$
Alpha translation	$A(z, 4Q) > L(n)$	$\alpha_J^Q \leq \alpha_Q + 1, \alpha_J^S \leq \alpha_Q + 2$
Cost bound	$f \leq km + 2nJ_k$	$f(m, n, L(n)) \leq (\alpha_Q + 3)m + 4n$

A compact text form of the dependency DAG is:

source definitions

```
-> J hierarchy
-> basic J lemmas
-> threshold recurrence
-> Ackermann domination
-> R/Ackermann comparison
-> alpha consequences
-> cost bound
```

source shifting lemma

```
-> recurrence f(m,n,r) <= k m + 2n J_k(r)
-> cost bound
```

packet normalizations $L(n), Q(m, n)$

```
-> alpha consequences
-> cost bound
```

APPENDIX B. FINITE SANITY CHECK

The script `scripts/sanity_check_j_thresholds.py` is a finite off-by-one sanity check. It computes small values of the packet J , R , and capped A definitions, checks monotonicity in small ranges, checks small instances of the threshold step, and checks the smallest published comparison instance

$R_2(1) \geq A(1, 4)$. It is not a proof and is not used as a dependency in the argument above.

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